

# Zachary Carlin

## Senior Software Engineer | Frontend & Full-Stack TypeScript

Tampa, Florida · Remote US · 727 505 9511 · [zacarlin@gmail.com](mailto:zacarlin@gmail.com) · [linkedin.com/in/zachary-carlin-85402a123](https://linkedin.com/in/zachary-carlin-85402a123) · [github.com/about7sharks](https://github.com/about7sharks)  
Authorized to work in the United States · No sponsorship required

---

### PROFESSIONAL SUMMARY

Frontend engineer who owns product from architecture to shipped software. Eight years building and modernizing data-rich web platforms in TypeScript, React, Next.js, and Node.js; currently the frontend technical lead for a B2B financial-data platform. Also design, build, and operate production systems end-to-end: a deployed customer-facing platform and private AI reliability infrastructure.

### TECHNICAL SKILLS

**Frontend:** TypeScript, JavaScript, React, Next.js, Vue, Nuxt, HTML, CSS, responsive design, server-side rendering, Figma implementation

**Backend and data:** Node.js, Python, REST APIs, GraphQL, SQL, Postgres, MongoDB, Firebase

**Delivery and quality:** Docker, Git, CI/CD, AWS, Google Cloud, Jest, test-driven development, ESLint, technical-debt controls

**Applied AI systems:** tool use, retrieval, deterministic evaluations, redaction, regression comparison, human approval, observability

### PROFESSIONAL EXPERIENCE

#### The Fly (Main Street Data) — Senior Software Engineer

Remote · July 2023–Present

- Frontend technical authority on a B2B financial-data platform — direction, delivery, and the build, from Figma through tested production.
- Built and operate the MCP server at [mcp.thefly.com](https://mcp.thefly.com), exposing The Fly as tools agents can call — production MCP tooling, not a demo.
- Built the product's AI chat widget end-to-end, including streaming responses, retrieval grounding, and human-gated behavior.
- Added automatic code review to the delivery flow, so every change ships with review and CI/CD gates before it can land.
- Improved the CI/CD pipeline — faster, more reliable builds and gated promotion — across the frontend and API surfaces.
- Modernized frontend quality controls (ESLint flat config, stricter checks, type-debt tracking) and led the Next.js pages-to-app-router migration with server-side rendering to cut initial-load latency.

#### Sonobi — Senior Software Engineer

Remote · July 2022–July 2023

- Built and maintained React applications across legacy class components, modern hooks, and asynchronous product flows.
- Replaced legacy jQuery workflows with maintainable React components.
- Integrated React features with PHP and Python services and improved GraphQL/Postgres-backed API paths.
- Contributed to Prebid.js and coordinated daily engineering delivery across the team.

#### Dogwood Logic — Software Engineer

Remote · February 2021–May 2022

- Implemented decentralized identifiers and verifiable credentials with Node.js and TypeScript.
- Built Vue.js interfaces for encrypted healthcare applications.
- Used test-driven development and realistic user-flow simulations to improve reliability.

#### Gulf Photonics — Software Engineer

Tampa, Florida · July 2018–August 2020

- Led software development for an IoT embedded sensor and its web dashboard.
- Built Node.js and Vue.js interfaces and APIs for sensor interaction.
- Managed MongoDB and Firebase data stores across mobile, desktop, Unix, and Windows environments.

### INDEPENDENT ENGINEERING

#### Z4C & AI Reliability Systems — Owner / Product Engineer / Operator

2024–Present

- Design, build, deploy, and operate a customer-facing platform (React, TypeScript, Cloudflare) end-to-end — timezone-aware intake, concurrency-safe storage, abuse protection, analytics, automated tests, production verification on every change. Live at [z4cllc.com](https://z4cllc.com).
- Build and operate private AI workflow and evaluation infrastructure: deterministic evaluations, regression comparison, redaction of persisted output, human-gated promotion with rollback awareness. Production agent tooling, not a demo.
- Own both systems end-to-end — architecture, code, deployment, and day-to-day operation.

### EDUCATION

Associate of Arts — St. Petersburg College, 2018